

PRINCIPLES OF PROGG LANG CS F301

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The First Week Quiz: Consciousness

Opened: Friday, 7 August 2026, 5:30 PM

Closed: Friday, 7 August 2026, 6:05 PM

We want you to start looking back at your own programming and software development habits, the activities you perform while programming and the thought process behind it.

- The task is to express the general process of comparison-based sorting in the following manner:
- Put succinctly, sorting is performed by repeatedly choosing at least one unique pair of two elements to compare in one iteration, and , depending upon the result of the comparisons, moving at least one of the compared elements closer to its right place by at least 1 index position in that iteration.
- Thus each iteration reduces at least one inversion (two elements in inverse positions by the ordering criterion). Thus, 1 2 **4 3** 5 has exactly one inversion (shown in **bold**) and swapping 4 and 3 removes it.
- Take any comparison-based sorting algorithm and its implementation in C or C++ or Java, according to your preference. (Bucket sort, radix sort, hashing to sort, etc. are not comparison-based. BubbleSort, QuickSort, InsertionSort, SelectionSort, MergeSort, HeapSort, etc. are.) Take the implementation ready-made from any source, including your own code.
- Using an existing implementation of a sorting algorithm, develop a wrapper program that takes array input, passes it on to this implementation, and depicts the sorting process showing the reduction of inversions in each iteration (something like showing the changing array, with the most recent swaps shown in bold).
- Most likely you will choose to modularise the original implementation such that the wrapper program can observe the iterations as input-output of unit steps, instead of inserting chaotic reporting actions in the sorting functions themselves.

Attempts allowed: 1

Time limit: 25 mins

Your attempts

Attempt 1

Status	Finished
Started	Friday, 7 August 2026, 5:32 PM
Completed	Friday, 7 August 2026, 5:52 PM
Duration	19 mins 47 secs

Review not permitted

No more attempts are allowed

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CS F301

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